

21ECP401L–MAJOR PROJECT

**AN INTELLIGENT MULTI-SENSOR WINDSHIELD
MONITORING FRAMEWORK WITH UWB ACTUATOR
CONTROL AND LORA-BASED V2V HAZARD
COMMUNICATION**

A PROJECT REPORT

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In partial fulfillment of the requirements for

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BONAFIDE CERTIFICATE

We hereby certify that the project report entitled “**An Intelligent Multi-Sensor Windshield Monitoring Framework with UWB Actuator Control and LoRa-Based V2V Hazard Communication**”, submitted under the course **21ECP401L – Major Project**, represents the genuine work undertaken by the following students:

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DECLARATION

We affirm that the Major Project titled **“An Intelligent Multi-Sensor Windshield Monitoring Framework with UWB Actuator Control and LoRa-Based V2V Hazard Communication”**, prepared for the award of the Bachelor of Technology degree, represents our authentic collaborative effort. We further confirm that this dissertation has not previously served as the basis for conferring any degree, diploma, fellowship, associateship, or comparable academic distinction.

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Abstract

Adverse weather phenomena, particularly precipitation and dense fog, deteriorate driving comfort while substantially elevating accident risk by curtailing the driver's field of view. Reliance on manual operation of wipers and defoggers under such circumstances introduces cognitive load and slows down corrective responses, both of which are critical safety factors. To mitigate these issues, this work proposes a wirelessly coordinated, intelligent windshield management platform that additionally supports vehicle-to-vehicle hazard signalling.

The proposed framework couples a rain detection sensor and a humidity-temperature module with an ESP32 controller, which engages the wiper mechanism automatically once rainfall is registered, and modulates the sweep rate in proportion to precipitation intensity. Actuator commands are delivered wirelessly using Ultra-Wideband (UWB) links via the RYUW122 module, eliminating the need for conventional point-to-point wiring inside the cabin.

Fog formation is inferred from sustained humidity readings, after which the windshield's heating element is energised to clear condensation and restore forward visibility. To safeguard the heating circuit, threshold limits and an automatic cut-off routine are embedded in the firmware to prevent thermal runaway.

For cooperative awareness across road users, the ESP-NOW protocol is employed to broadcast situational alerts such as "rain ahead" or "visibility reduced by fog" to neighbouring vehicles, allowing other drivers to anticipate the changing conditions before encountering them directly.

Sustainable Development Goal:

By advancing road safety and contributing to smarter transportation infrastructure, the proposed framework aligns with several United Nations SDGs, namely **SDG 3 (Good Health and Well-being)**, **SDG 9 (Industry, Innovation, and Infrastructure)**, and **SDG 11 (Sustainable Cities and Communities)**.



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List of Symbols and Abbreviations

Abbreviations

Abbreviation	Description
ESP32	Espressif 32-bit Microcontroller
UWB	Ultra-Wideband
LoRa	Long Range Communication
V2V	Vehicle-to-Vehicle Communication
IR	Infrared
DHT	Digital Humidity and Temperature Sensor
OLED	Organic Light Emitting Diode
TFT	Thin Film Transistor Display
GPIO	General Purpose Input/Output
UART	Universal Asynchronous Receiver Transmitter
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit
RSSI	Received Signal Strength Indicator

Symbols

Symbol	Description
°C	Temperature in degree Celsius
%	Percentage (Humidity)
V	Voltage
A	Current
ms	Milliseconds
GHz	Gigahertz
m	Meter (Distance)

Chapter 1

Introduction

1.1 Background

Road transportation safety remains a cornerstone concern for modern mobility, and meteorological factors such as rainfall and fog routinely undermine it by impairing how clearly the driver perceives the road, lengthening braking distances, and slowing reaction time. Statistical evidence consistently links a sizeable share of vehicular collisions to degraded weather, and the risk grows sharply on high-speed corridors as well as in congested urban traffic. When visibility falls, the driver must intervene with wipers and defogging controls, yet each such intervention competes with the cognitive resources needed for safe driving and lengthens the time taken to react during emergencies. These realities motivate the development of intelligent in-vehicle systems capable of perceiving the surrounding environment and restoring visibility autonomously.

1.2 Problem Statement

In conventional vehicles, the management of windshield clarity continues to depend on the driver's manual interaction with wiper switches and defogging controls. Such conventional arrangements are not adaptive, lack on-board intelligence, and provide no mechanism to share information with other vehicles. The principal weaknesses of the prevailing approach can be summarised as follows:

- Latency in engaging wipers and defogging once visibility deteriorates
- Additional cognitive demand placed on the driver
- No advance awareness of hazardous conditions further along the route
- Bulky wiring harnesses linking sensors with their corresponding actuators
- Absence of any cooperative safety exchange between neighbouring vehicles

1.3 Objectives

The principal aim of this work is to engineer a Smart Wireless Windshield Management and Vehicle-to-Vehicle Alerting System that strengthens both the safety margin and the overall driving experience. The defining capabilities targeted by the proposed system include:

- Automated identification of rain and fog through dedicated sensing elements
- Dynamic regulation of wiper sweep speed in response to rainfall intensity
- Automatic engagement of the defogging mechanism upon fog detection
- Substitution of conventional wiring with wireless data transport
- Real-time dissemination of hazard notifications to surrounding vehicles via V2V messaging

1.4 Scope of the Project

This project encompasses the design and prototyping of an intelligent windshield monitoring platform that supports drivers by recognising environmental disturbances which compromise forward visibility. The framework brings together the disciplines of sensing, wireless communication, and electromechanical actuation within a single coordinated solution.

The platform deploys a complementary set of sensors to capture different visibility-related parameters from the immediate environment of the windshield. The acquired readings are interpreted by an ESP32 microcontroller, which then dispatches actuation instructions over a low-latency Ultra-Wideband channel. By replacing internal wiring with a UWB link between the sensor cluster and the actuator cluster, the design achieves a cleaner cabin layout while preserving deterministic response timing.

Beyond intra-vehicle coordination, the framework integrates inter-vehicle messaging based on LoRa technology, allowing nearby vehicles to be notified of the detected hazards. The current effort is, however, intentionally bounded to the realisation of a working prototype rather than full automotive-grade deployment.

1.5 Organization of the Report

The remainder of this dissertation has been arranged systematically in line with the Major Project Report guidelines issued by SRM Institute of Science and Technology. The material is partitioned into five principal chapters, each addressing a distinct facet of the project.

Chapter 2

Literature Review

2.1 Existing Work

A considerable body of automotive research has concentrated on improving driver visibility and lowering the probability of weather-induced collisions. Production cars now routinely incorporate rain-triggered wipers and electrically heated defoggers [9], yet most of these features remain bound to single-input thresholding, which restricts their usefulness in nuanced operating conditions.

More recent contributions in vehicular sensing, including radar-based perception and in-cabin monitoring, have widened the perceptual envelope available to the vehicle and enabled richer safety functions [7]. Parallel work on intelligent control architectures has also examined how automation can shorten the loop between perception and actuation [6].

Sensor-fusion strategies have likewise been investigated as a route to greater situational awareness, since the joint interpretation of multiple modalities can compensate for the weaknesses of any individual sensor [8]. Such methods, however, have largely been deployed in premium autonomous platforms where computational headroom is generous.

In the wireless domain, several technologies—including DSRC, Wi-Fi, and LoRa—have been examined for vehicle-to-vehicle exchange [9, 10]. LoRa stands out for its long reach and low power demand, qualities that suit hazard-broadcast applications, although its non-trivial latency makes it ill-suited for time-critical control loops.

Ultra-Wideband (UWB) communication has therefore attracted growing interest because it combines very low latency with high spatial precision, both of which are valuable for real-time actuator coordination and proximity sensing inside vehicles [1, 2]. Studies on UWB antenna engineering and its integration into automotive communication stacks have further matured the technology [3].

Wireless control schemes and embedded automation techniques have additionally been adopted across several engineering disciplines to streamline operation and improve flexibility [5]. In a similar spirit, low-cost embedded monitoring deployments in unrelated domains have demonstrated that scalable sensing solutions can be realised on modest hardware budgets [4].

In spite of these contributions, the literature still lacks a cohesive framework that si-

multaneously exploits multi-sensor perception, real-time wireless actuation, and dual-range communication for both intra- and inter-vehicle messaging. The specific scenario of an abruptly obstructed windshield has likewise received limited attention. The present project seeks to close these gaps by uniting multi-modal sensing with UWB-driven actuation and LoRa-based V2V signalling.

2.2 Gap Analysis

While automotive safety research has progressed substantially, several gaps continue to limit the practical effectiveness of current solutions, particularly when visibility deteriorates rapidly or fluctuates in complex ways during precipitation or fog episodes. The conventional rain-sensing wiper or stand-alone defogger functions in isolation, drawing on a single sensor input, and is therefore poorly placed to interpret combined or evolving environmental cues.

Most published multi-sensor fusion work targets premium autonomous vehicles where powerful computing platforms are available—an assumption that is incompatible with affordable, broadly deployable safety modules. Equally importantly, existing systems rarely consider the case in which the windshield becomes suddenly opaque due to external matter such as cloth, leaves, or debris; this corner case constitutes a real safety concern that current solutions overlook.

On the communications front, LoRa has been adopted for V2V messaging owing to its long-range characteristics, yet its higher latency rules it out for tight control loops. Conversely, Ultra-Wideband offers the low-latency channel needed for in-cabin actuation but has not yet been routinely integrated into mainstream automotive safety designs. The absence of a unified architecture that brings sensors, actuators, and dual-mode communication together remains a defining limitation of the field.

Table 2.1: Communication Technology Comparison

Parameter	UWB	LoRa	Wi-Fi
Range	Short	Long	Medium
Latency	Very Low	High	Medium
Data Rate	High	Low	High
Power Consumption	Low	Very Low	High
Application	Actuator Control	V2V Alerts	Video Streaming

2.3 Summary

The work outlined here proposes a novel multi-sensor windshield monitoring scheme that supports drivers in maintaining safety when visibility is degraded by external influences such as rain or fog. A coordinated combination of rain, humidity, and infrared sensors continuously observes the immediate environment, while an ESP32 embedded platform interprets the readings in real time and engages the wipers and defoggers automatically, thereby cutting both driver workload and reaction delay.

To strengthen system performance, intra-vehicle coordination relies on the low-latency UWB protocol that links the sensing node with the actuator node. Long-range broadcasting of detected hazards to surrounding vehicles is handled separately by a LoRa link, while the ESP32-CAM module supplies supplementary visual feedback to the driver whenever it is required.

Experimental observations indicate that the proposed arrangement responds promptly and behaves consistently across the tested weather scenarios, presenting an affordable and effective option for augmenting present-day automotive safety systems.

Chapter 3

Methodology / System Design

3.1 Architecture

The architecture adopted in this project follows a modular and distributed organisation built around three cooperating subsystems: a sensing node, an actuation node, and a communication node. Acting in concert, these subsystems deliver continuous monitoring, on-the-fly decision making, and timely intervention against visibility-related hazards encountered during driving.

The sensing node is tasked with collecting environmental measurements through a complementary set of transducers—a rain sensor, a DHT22 temperature and humidity sensor, and an infrared (IR) sensor for detecting unexpected obstructions on the windshield. All three transducers are routed to an ESP32 microcontroller, which interprets the incoming streams in real time to recognise rainfall, fog formation, and physical blockages. Whenever the readings cross pre-set thresholds, the ESP32 issues the appropriate actuation command.

The actuation node is implemented as a physically separate module reachable over an Ultra-Wideband (UWB) link. Commands generated by the sensing ESP32 are dispatched wirelessly to the actuator ESP32 through this UWB channel, ensuring deterministic and low-latency delivery. At the receiving end, the actuator drives a servo motor that emulates the wiper sweep and switches a relay that engages a defogging element such as a small fan or heating coil. By avoiding internal wiring between sensing and actuation, the architecture reduces installation complexity and offers greater design flexibility within the cabin.

The communication node enlarges the system's reach beyond the host vehicle. A LoRa transceiver broadcasts hazard messages in real time to vehicles travelling nearby, while an ESP32-CAM module supplies a live video stream—rendered on a TFT display—whenever the windshield becomes obstructed. Together, these modules unite sensing, actuation, and communication into a coherent platform that improves driver awareness as well as overall safety.

The mapping between detected environmental conditions, the responsible sensor, and the resulting system action is consolidated in Table 3.1.

Table 3.1: Sensor–Action Mapping

Condition Detected	Sensor Used	Action Taken
Rain	Rain Sensor	Wiper activated
High Humidity (Fog)	DHT22	Defogger activated
Obstruction	IR Sensor	Camera activated + alert
Low Visibility	Combined sensors	LoRa alert transmitted

3.2 Tools and Technologies Used

Realising the proposed smart windshield monitoring platform requires the joint use of carefully selected hardware modules, software environments, and communication standards so that sensing, actuation, and data exchange can occur reliably and in real time.

- Hardware Components

The ESP32 microcontroller anchors the platform, having been chosen for its strong computational capacity and the breadth of its peripheral interfaces. Environmental measurements are obtained from a rain sensor for precipitation, a DHT22 module for ambient temperature and humidity, and an infrared sensor for sensing obstructions on the windshield. On the actuation side, a servo motor reproduces the wiper sweep, while a relay-controlled DC fan or heating coil mimics the defogging mechanism. An ESP32-CAM module captures live imagery whenever visibility is interrupted, and OLED and TFT displays surface system status as well as alerts to the driver. The full hardware inventory along with the corresponding specifications is consolidated in Table 3.2.

- Communication Technologies

Intra-vehicle wireless coordination between the sensing and actuation nodes is achieved with the RYUW122 Ultra-Wideband module, which provides the low-latency channel needed for time-sensitive control. Long-range messaging to neighbouring vehicles is handled through the RYLR998 LoRa module, while the ESP32-CAM relies on Wi-Fi for live video streaming.

- Software Tools

Firmware for the ESP32 family, including the ESP32-CAM, is developed in the Arduino IDE using embedded C/C++. This codebase implements the sensor interfaces, the decision logic, and the various communication protocols, supported by the relevant device libraries that simplify integration with the displays, sensors, and radio modules.

Table 3.2: Hardware Components and Specifications

Component	Model	Function
Microcontroller	ESP32 DevKit V1	Main processing unit
Camera Module	ESP32-CAM	Real-time video capture
Rain Sensor	FC-37	Rain detection
Temperature & Humidity	DHT22	Fog detection
IR Sensor	IR Module	Obstruction detection
LoRa Module	RYLR998	V2V communication
UWB Module	RYUW122	In-vehicle communication
Relay Module	4-Channel Relay	Actuator control
Servo Motor	SG90	Wiper simulation
DC Fan	—	Defogger simulation
OLED Display	SSD1306	Status display
TFT Display	—	Video output

3.3 SYSTEM BLOCK DIAGRAM

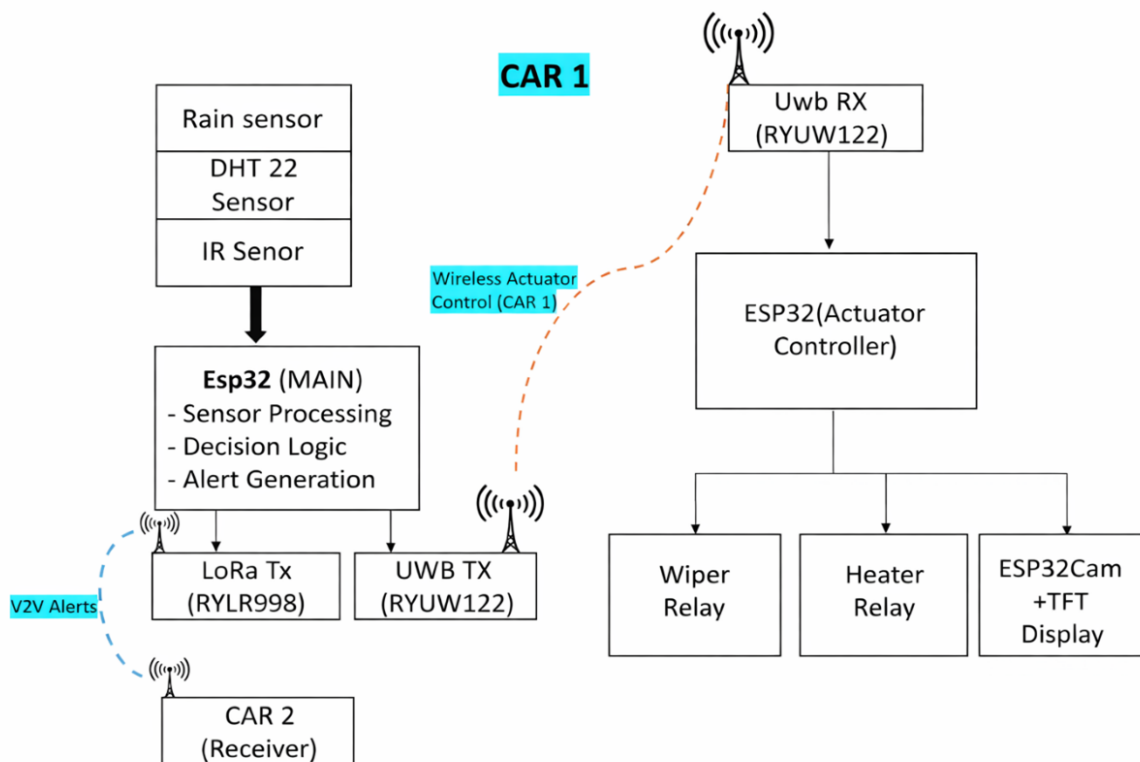


Figure 3.1: System Design and Architecture of CAR 1.

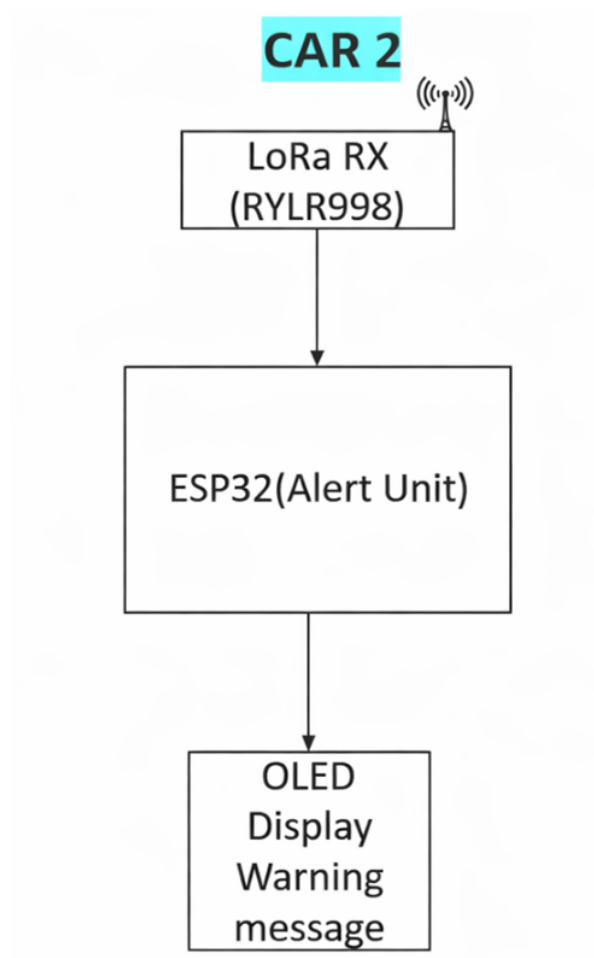


Figure 3.2: System Design and Architecture of CAR 2.

Chapter 4

Implementation and Results

4.1 System Implementation

The smart windshield monitoring platform was constructed using a modular layout comprising three cooperating units: a sensing node, an actuation node, and a communication node. At the sensing end, an ESP32 microcontroller was wired to a rain sensor, a DHT22 temperature–humidity sensor, and an infrared sensor for detecting obstructions, allowing continuous observation of the surrounding environment and immediate availability of fresh data to the controller.

The control logic is built around predefined thresholds applied to the sensor stream. When precipitation is registered, an actuation command is generated to engage the wiper. Likewise, once the humidity reading exceeds the limit set for fog conditions, the defogging path—driven by a relay that switches a heating coil or fan—is energised. For unexpected obstructions, the IR sensor flags the sudden blockage and triggers an alert together with the visual assistance feed produced by the ESP32-CAM module.

In-cabin wireless coordination relies on RYUW122 Ultra-Wideband modules, which carry the actuation commands from the sensing ESP32 to the actuator ESP32 with very low latency. A second ESP32 on the actuation side decodes the incoming UWB messages and drives the servo motor that mimics the wiper sweep, alongside the relays that govern the defogging path. For inter-vehicle communication, an RYLR998 LoRa module broadcasts hazard notifications such as rain, fog, or low-visibility conditions to vehicles in the vicinity.

Table 4.1: System Performance Evaluation

Parameter	Observed Value
Response Time	< 0.5 seconds
Rain Detection Accuracy	High
Fog Detection Threshold	$\geq 85\%$ Humidity
UWB Communication Delay	~ 10 ms
LoRa Transmission Range	Up to several km
System Stability	Consistent

4.2 Experimental Setup

The platform was exercised under controlled bench-top conditions in order to characterise its behaviour. Rainfall was emulated by depositing water droplets on the surface of the rain sensor, whereas fog episodes were reproduced by elevating the local humidity around the DHT22 module. Obstruction events, in turn, were generated by interposing items such as fabric and paper in front of the IR sensor.

Two ESP32 boards joined through the UWB link were employed to verify wireless actuator coordination, as illustrated by the sensing node in Fig. 4.1 and the actuator node in Fig. 4.2. Independently, LoRa modules attached to additional ESP32 boards were configured to demonstrate the V2V message exchange, while the ESP32-CAM module reproduced the visual-assistance scenario when forward vision was blocked.

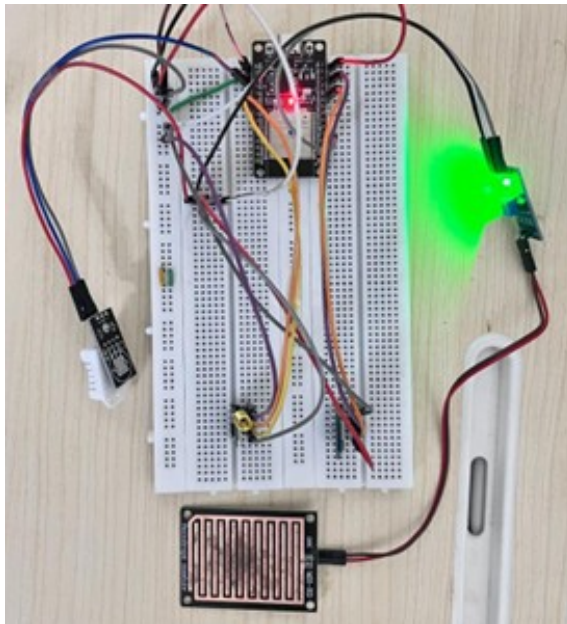


Figure 4.1: EXPERIMENTAL SETUP: CAR 1 (Sensing Node).

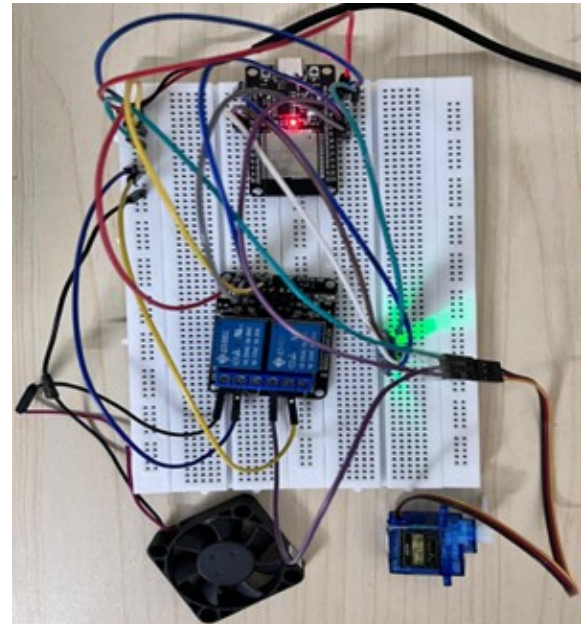


Figure 4.2: EXPERIMENTAL SETUP: CAR 1 (Actuator Node).

4.3 Results and Discussions

Across all evaluated scenarios, the prototype delivered dependable and repeatable behaviour. The rain sensor reliably registered moisture events and the corresponding actuation triggered the servo motor that simulates the wiper sweep. The DHT22 module produced consistent temperature and humidity readings, and once the humidity crossed the fog threshold the defogging path was engaged without false triggering.

Wireless coordination over UWB between the sensing and actuator nodes was established successfully, providing low-latency control of the actuators. Measured end-to-end response

time remained below 0.5 seconds, which is short enough to keep pace with rapidly changing environmental conditions.

LoRa-based broadcasting of hazard notifications also performed reliably across repeated trials. Distinct alert classes—rain, fog, and combined rain-and-fog—were generated and successfully received by the partner unit, with the link remaining stable throughout testing.

The visual outputs produced by the system under different conditions are reproduced in Fig. 4.3–Fig. 4.6: Fig. 4.3 captures the default clear-weather state, Fig. 4.4 corresponds to a fog-only alert, Fig. 4.5 shows the rain-only alert, and Fig. 4.6 depicts the combined rain-and-fog notification.

The obstruction-detection path likewise succeeded in flagging windshield blockages, and the ESP32-CAM module supplied real-time imagery to the driver, confirming the practicality of providing visual assistance during sudden visibility loss.



Figure 4.3: Default (Clear) Condition Display.



Figure 4.4: Fog Alert Display.



Figure 4.5: Rain Alert Display.



Figure 4.6: Combined Rain and Fog Alert Display.

Chapter 5

Conclusion and Future Work

5.1 Summary

The intelligent smart windshield monitoring platform proposed in this work successfully illustrates how visibility-related challenges in driving environments can be detected and addressed in real time. By coupling rain, humidity, and infrared sensing with an ESP32-class microcontroller, the system reliably distinguishes between rainfall, fog formation, and unexpected windshield blockages. These observations are then translated into automatic actuation of the wiper and defogger, thereby easing the cognitive burden on the driver and shortening reaction time.

The choice of Ultra-Wideband communication enables responsive, low-latency exchange between the in-cabin sensing and actuation modules, while the LoRa link extends situational awareness to surrounding vehicles through hazard alerts. The integrated camera module further reinforces the platform's usefulness during sudden visibility loss. Experimental trials confirm that the system delivers prompt and stable behaviour across the range of conditions considered.

5.2 Key Outcomes

- Realisation of a multi-sensor fusion stack capable of distinguishing rain, fog, and windshield obstruction events
- Automatic ESP32-driven control of wiper and defogger paths in real time
- Dependable UWB-based wireless link providing low-latency actuator coordination inside the vehicle
- Robust LoRa-based V2V exchange that conveys hazard alerts under different weather conditions
- Camera-assisted visibility support tailored to sudden obstruction scenarios

- End-to-end response time held below 0.5 seconds, comfortably outperforming manual operation
- A low-cost, scalable, and energy-conscious prototype that is well aligned with practical automotive safety needs

5.3 Limitations

While the prototype performs effectively in a controlled environment, a number of limitations should be acknowledged. The current build has not yet been validated within an operational vehicle, where factors such as fluctuating weather, mechanical vibration, and ambient electromagnetic interference can influence performance. Fog inference is presently based on a simple humidity threshold rather than on a more accurate dew-point computation, which can introduce minor classification errors.

In addition, the ESP32-CAM module supports only basic video streaming and has limited resolution as well as on-board processing capability. The LoRa link is appropriate for short hazard messages but cannot accommodate high-bandwidth payloads such as live video. Power management and overall system integration likewise remain at a prototype level and would require further refinement prior to field deployment.

5.4 Scope for Future Work

Several enhancements could meaningfully extend both the performance and applicability of the platform. Incorporating advanced image-processing pipelines and machine-learning models would permit more accurate obstruction classification and predictive estimation of visibility trends. The camera subsystem itself could be upgraded to a higher-resolution unit equipped with on-device inference for improved driver assistance.

Coupling the platform with established in-vehicle networks such as CAN bus would simplify integration with modern automotive electronics. Adding GPS support would further enable location-aware hazard alerts to be transmitted via the LoRa link. Finally, refinements in hardware design—miniaturisation, improved power management, and ruggedisation—would move the prototype closer to a deployable, commercially viable automotive safety solution.

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Appendix

Appendix A: Sensor Unit Code (ESP32 – Vehicle 1)

The listing below corresponds to the sensing node firmware, which is responsible for observing rainfall, humidity-derived fog conditions, and windshield obstruction. The ESP32 processes the acquired readings and transmits the corresponding control instructions over the UWB and LoRa links.

Sensor Unit Code (Rain + DHT22 + UWB + LoRa)

```
#include "DHT.h"

// Pin Definitions based on your ESP32 Pinout
#define DHTPIN 15
#define DHTTYPE DHT22
#define RAIN_PIN 34

// Thresholds - Adjusted to 30.0 as requested
float TEMP_THRESHOLD = 30.0;
int RAIN_THRESHOLD = 2500;    // Lower value means more water detected

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);        // Debug to Computer

  // RYLR998 LoRa on Serial2 (GPIO 16=RX, 17=TX)
  Serial2.begin(115200, SERIAL_8N1, 16, 17);

  // RYUW122 UWB on Serial1 (GPIO 13=RX, 14=TX)
  // Note: TX on ESP32 goes to RX on Module
  Serial1.begin(115200, SERIAL_8N1, 13, 14);
```

```

dht.begin();
pinMode(RAIN_PIN, INPUT);

Serial.println("--- Brain Node Initialized ---");
}

void loop() {
  float temp = dht.readTemperature();
  int rainVal = analogRead(RAIN_PIN);

  // Check if sensor reading failed
  if (isnan(temp)) {
    Serial.println("Failed to read from DHT sensor!");
    temp = 0;
  }

  String status = "CLEAR";

  // Logic for detection
  bool isRaining = (rainVal < RAIN_THRESHOLD);
  bool isFoggy = (temp > TEMP_THRESHOLD);

  if (isRaining && isFoggy) status = "RAIN_FOG";
  else if (isRaining) status = "RAIN";
  else if (isFoggy) status = "FOG";
  else status = "CLEAR";

  // --- 1. SEND VIA UWB (Internal Actuator) ---
  String uwbCmd = "AT+ANCHOR_SEND=SLAVE001," + String(status.length())
  + "," + status + "\r\n";
  Serial1.print(uwbCmd);
  Serial1.flush();

  // --- 2. SEND VIA LORA (Rear Vehicle Car-2) ---
  String loraCmd = "AT+SEND=2," + String(status.length())
  + "," + status + "\r\n";
  Serial2.print(loraCmd);
  Serial2.flush();
}

```

```

// Local Debugging
Serial.print("Temp: "); Serial.print(temp);
Serial.print("C | Rain Value: "); Serial.print(rainVal);
Serial.print(" | Sending Status: "); Serial.println(status);

delay(2000);
}

```

Appendix B: Actuator Unit Code (ESP32 – Vehicle 1)

The following listing implements the actuation node, which receives commands over the UWB link and drives the wiper servo, the defogger fan, and the relay-controlled outputs accordingly.

```

#include

#define WIPER_RELAY_PIN 27
#define HEATER_RELAY_PIN 25
#define SERVO_PIN 26

Servo wiperServo;

bool wiperActive = false;
int servoPos = 30;
int direction = 1;

const int MIN_ANGLE = 0;
const int MAX_ANGLE = 120;

const int SPEED_DELAY = 4;
const int STEP_SIZE = 8;

unsigned long lastMove = 0;

void setup() {
  Serial.begin(115200);
  Serial2.begin(115200, SERIAL_8N1, 16, 17);

```

```

digitalWrite(WIPER_RELAY_PIN, HIGH);
digitalWrite(HEATER_RELAY_PIN, HIGH);
pinMode(WIPER_RELAY_PIN, OUTPUT);
pinMode(HEATER_RELAY_PIN, OUTPUT);

wiperServo.attach(SERVO_PIN);
wiperServo.write(MIN_ANGLE);

Serial.println("Actuator Listening for UWB Alerts...");
}

void loop() {
  if (Serial2.available()) {
    String incoming = Serial2.readString();
    incoming.trim();

    Serial.print("RAW DATA: ");
    Serial.println(incoming);

    if (incoming.indexOf("RAIN_FOG") != -1) {
      controlHardware(true, true);
    }
    else if (incoming.indexOf("RAIN") != -1) {
      controlHardware(true, false);
    }
    else if (incoming.indexOf("FOG") != -1) {
      controlHardware(false, true);
    }
    else if (incoming.indexOf("CLEAR") != -1) {
      controlHardware(false, false);
    }
  }

  if (wiperActive) {
    sweepWiper();
  }
}

```

```

void controlHardware(bool wiper, bool heater) {
    if (wiper) {
        digitalWrite(WIPER_RELAY_PIN, LOW);
        wiperActive = true;
    }
    else {
        digitalWrite(WIPER_RELAY_PIN, HIGH);
        wiperActive = false;
        servoPos = MIN_ANGLE;
        wiperServo.write(MIN_ANGLE);
    }

    digitalWrite(HEATER_RELAY_PIN, heater ? LOW : HIGH);
}

void sweepWiper() {
    if (millis() - lastMove > SPEED_DELAY) {
        servoPos += direction * STEP_SIZE;

        if (servoPos >= MAX_ANGLE) {
            servoPos = MAX_ANGLE;
            direction = -1;
        }

        if (servoPos <= MIN_ANGLE) {
            servoPos = MIN_ANGLE;
            direction = 1;
            delay(80);
        }

        wiperServo.write(servoPos);
        lastMove = millis();
    }
}

```

Appendix C: Vehicle 2 Code (LoRa Receiver System)

The listing below implements the second vehicle's receiver, which captures incoming hazard messages over the LoRa link and renders the corresponding warnings on the OLED/TFT display.

Vehicle 2 Code (LoRa Receiver + Display)

```
#include
#include
#include

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
  Serial.begin(115200);
  Serial2.begin(115200, SERIAL_8N1, 16, 17);

  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    for(;;);
  }

  showBootScreen();
}

void loop() {
  if (Serial2.available()) {
    String incoming = Serial2.readStringUntil('\n');

    display.clearDisplay();
    drawHeader();

    if (incoming.indexOf("FAST") >= 0) {
      drawAlertFrame("!! DANGER !!", "EXTREME RAIN");
    }
    else if (incoming.indexOf("RAIN_FOG") >= 0) {
      drawAlertFrame("WARNING", "VISIBILITY LOW");
    }
  }
}
```

```

    }
    else if (incoming.indexOf("RAIN") >= 0) {
        drawAlertFrame("CAUTION", "ROAD IS WET");
    }
    else if (incoming.indexOf("FOG") >= 0) {
        drawAlertFrame("CAUTION", "DENSE FOG");
    }
    else {
        drawNormalFrame("STATUS OK", "ROAD CLEAR");
    }

    display.display();
}
}

void drawHeader() {
    display.fillRect(0, 0, 128, 14, WHITE);
    display.setTextColor(BLACK);
    display.setTextSize(1);
    display.setCursor(35, 3);
    display.print("V2V ACTIVE");
}

void drawAlertFrame(String title, String msg) {
    display.setTextColor(WHITE);
    display.drawRect(0, 18, 128, 46, WHITE);

    display.setTextSize(2);
    int16_t x1, y1; uint16_t w, h;
    display.getTextBounds(title, 0, 0, &x1, &y1, &w, &h);
    display.setCursor((128 - w) / 2, 25);
    display.println(title);

    display.setTextSize(1);
    display.getTextBounds(msg, 0, 0, &x1, &y1, &w, &h);
    display.setCursor((128 - w) / 2, 48);
    display.println(msg);
}

```

```

void drawNormalFrame(String title, String msg) {
    display.setTextColor(WHITE);
    display.setTextSize(1);
    display.setCursor(38, 25);
    display.println(title);

    display.setTextSize(2);
    int16_t x1, y1; uint16_t w, h;
    display.getTextBounds(msg, 0, 0, &x1, &y1, &w, &h);
    display.setCursor((128 - w) / 2, 42);
    display.println(msg);
}

void showBootScreen() {
    display.clearDisplay();
    display.drawRoundRect(10, 10, 108, 44, 8, WHITE);
    display.setTextSize(2);
    display.setCursor(25, 25);
    display.println("SMART");
    display.display();
    delay(1000);
}

```

Appendix D: Circuit Connections

The tables that follow document the complete pin assignments and wiring relationships for the three ESP32 nodes used in the prototype, namely the sensing node, the actuator node, and the receiver node.

Table D.1: ESP32-1 (Sensing Node – Front Vehicle)

Component	Component Pin	ESP32 Pin
Rain Sensor	AO (Analog)	GPIO34
DHT22	Data	GPIO15
RYLR998 (LoRa)	TX	GPIO16 (RX2)
RYLR998 (LoRa)	RX	GPIO17 (TX2)
RYUW122 (UWB)	TX	GPIO13
RYUW122 (UWB)	RX	GPIO14

Table D.2: ESP32-2 (Actuator Node – Internal Vehicle)

Component	Component Pin	ESP32 Pin
RYUW122 (UWB)	TX	GPIO16 (RX2)
RYUW122 (UWB)	RX	GPIO17 (TX2)
Wiper Relay	IN	GPIO27
Heater Relay	IN	GPIO25
Servo Motor	Signal	GPIO26

Table D.3: ESP32-3 (Receiver Node – Rear Vehicle)

Component	Component Pin	ESP32 Pin
RYLR998 (LoRa)	TX	GPIO16 (RX2)
RYLR998 (LoRa)	RX	GPIO17 (TX2)
OLED Display	SDA	GPIO21
OLED Display	SCL	GPIO22

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PUBLICATION DETAILS

The presented research proposal on “An Intelligent Multi-Sensor Windshield Monitoring Framework with UWB Actuator Control and LoRa-Based V2V Hazard Communication” was successfully presented at **2nd International Green Energy Mobility Conference 2026 (IGEM 2026)**, which is one of the recognized international conferences dedicated to sustainable and intelligent means of mobility. The recognition of this proposal came in the form of a Certificate of Appreciation as this research is relevant to future developments in the field of vehicle safety. According to this official certificate, the researchers gave an oral presentation at this conference that was held on 9-10 April 2026 organized by the Department of Automobile Engineering, SRM Institute of Science and Technology, Kattankulathur, Tamil Nadu, India.

